Project Design Phase

Solution Architecture

Date	17 February 2026
Team ID	LTVIP2026TMIDS37778
Project Name	Laptop Request Catalog Item

Solution Architecture:

Solution Architecture – Laptop Request Catalog Item
The solution architecture of the **Laptop Request Catalog Item** is designed to automate and standardize the laptop request process using the ServiceNow Zurich platform. The system integrates Service Catalog, workflow automation, UI policies, and role-based access control to ensure efficient request handling and asset management.

Key Goals of the Architecture:

• Automate the manual laptop request process.
• Implement structured approval workflow.
• Enable dynamic form behavior using UI Policies.
• Ensure secure role-based access.
• Improve transparency and efficiency in asset allocation

Architecture Layers and Components:

User Interface Layer (Service Portal / Service Catalog)

• Laptop Request item created under Hardware category.
• Initially displays only three main variables.
• Clean and user-friendly form design.
• Accessible via Service Catalog for all employees.

Variables Configuration

The catalog item includes structured variables such as:

• Laptop Type (Select Box)
• Required Configuration (RAM, Storage, etc.)
• Justification (Multi-line Text)
• Additional Accessories (Checkbox)
• Accessories Details (Multi-line Text / Select Box)

Variable Behavior:

• Accessories Details field is hidden by default.
• Becomes visible only when “Additional Accessories” checkbox is selected.
• Becomes mandatory when checkbox is selected.

UI Policy Implementation

UI Policies are used to control dynamic behavior of variables without scripting.

UI Policy Conditions:

• Condition: If "Additional Accessories" checkbox is checked (True).

UI Policy Actions:

• Set Accessories Details field → Visible = True
• Set Accessories Details field → Mandatory = True
• Set Accessories Details field → Read-only = False

When checkbox is unchecked:

• Accessories Details field → Hidden
• Accessories Details field → Not Mandatory

This ensures proper data validation and prevents incomplete submissions.

Application Logic Layer (Flow Designer / Workflow)

• On submission, request record is created.
• Approval task automatically routed to Manager.
• After approval, task assigned to IT Admin.
• Status updates automatically throughout lifecycle.

Request Flow:

Employee → Manager Approval → IT Fulfillment → Request Closure

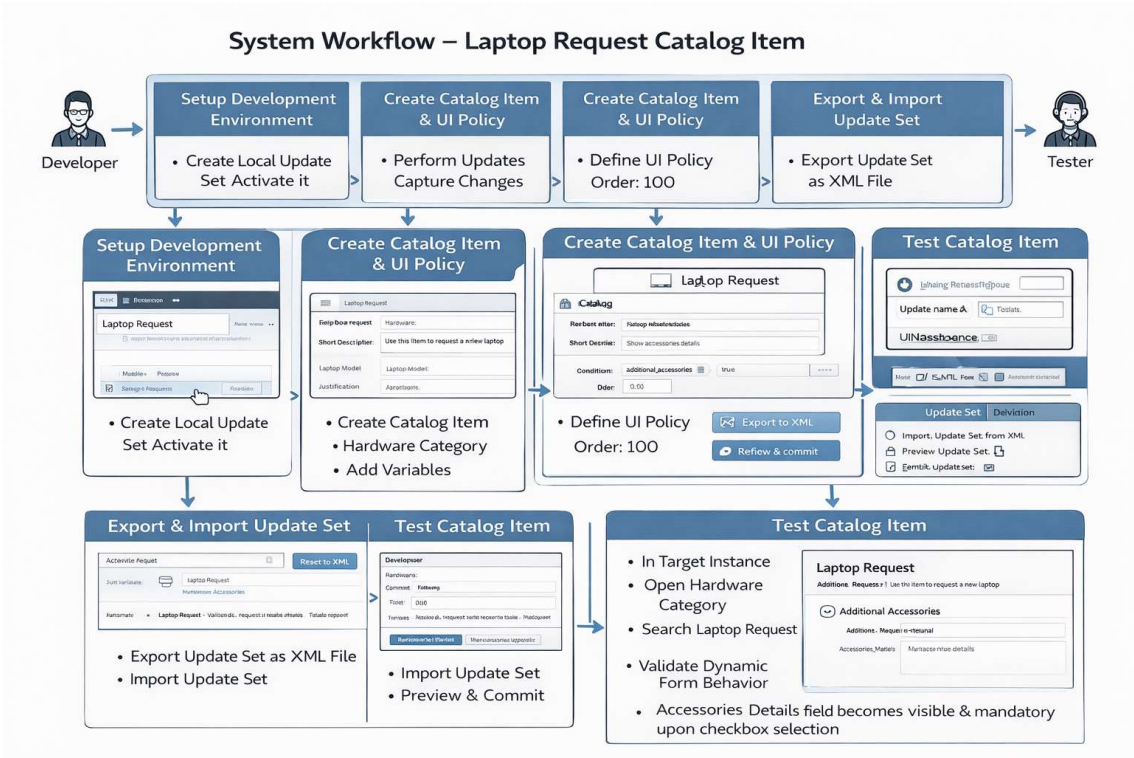
Role-Based Access Control (RBAC)

• Employee – Submit & track requests
• Manager – Approve/Reject requests
• IT Admin – Fulfill & close requests

Access Control Lists (ACLs) ensure secure record access.

Data Layer (ServiceNow Tables)

• sc_request – Stores request record
• sc_req_item – Stores requested item details
• sc_task – Stores approval and fulfillment tasks

Ensures centralized and traceable data management.

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